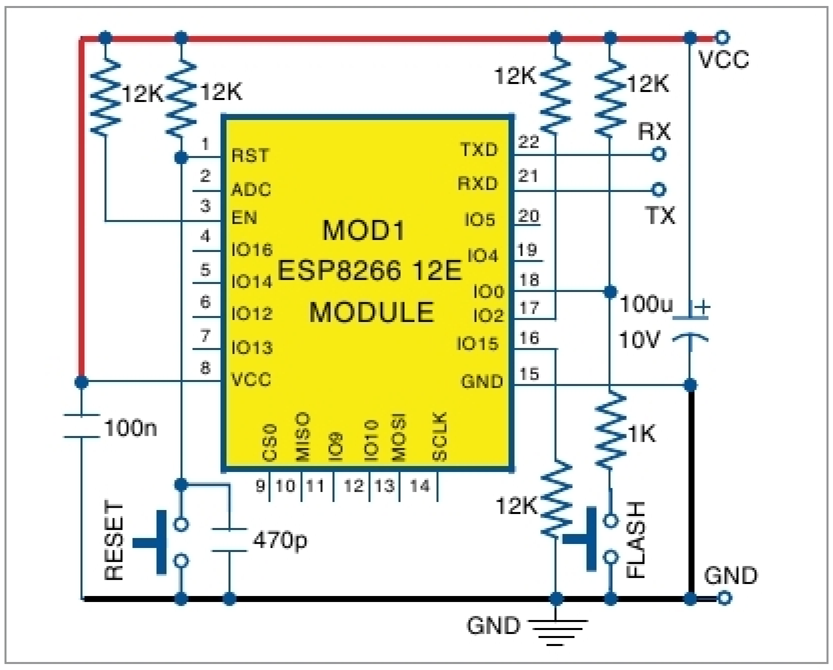

Fig. 4: Circuit diagram of manual ESP-12E/F programmer

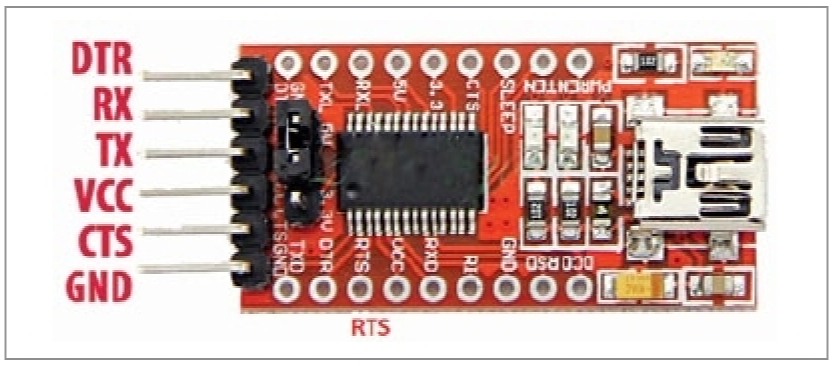

Fig. 5: FT232 USB-to-UART module

Different Boot Modes of ESP-12E/F

	GPIO 0	GPIO 2	GPIO 15	CH_PD (En)
Programming mode	0 Low	1 High	0 Low	1 High
Flash start-up (normal mode)	1 High	1 High	0 Low	1 High
SD card boot-up	0 Low	0 Low	1 High	1 High

like Arduino IDE, with ESP8266 Board Manager and supporting libraries, or the newer PlatformIO IDE, for writing the software.

1. Software code is flashed into ESP8266 chip using the built-in serial TTL-to-USB adaptor (sometimes called USB-to-UART bridge). For flashing any new software, ESP8266 chip must be rebooted in flash mode, and USB signals from the computer must be converted to serial UART signals and transferred to the chip using Rx0 and Tx0 pins (GPIO3 and GPIO1).

2. On NodeMCU board, support circuitry automatically handles all these things in a seamless manner. This makes NodeMCU or similar board, like WemosD1 Mini, popular among hobbyists. Note that, Reset and Flash buttons are also built onto NodeMCU board to help in case auto-loading fails.

Deployment/production stage

During this stage only ESP8266 chip is required in the final product.

The issue now is to figure out how to use ESP8266-12E/F (or ESP-12E/F) chip and still be able to load the code and develop the software without NodeMCU or similar board. For that, first we need to know the physical size, working, booting and flashing process of ESP-12E/F chip in detail.

ESP-12E/F module. External size of the module is 16mm x 24mm x 3mm, as shown in Fig. 2. It has 4MB SPI flash and an onboard PCB antenna. Pin pitch is 2mm.

If a general-purpose PCB, which has a pin spacing of 2.54mm, is used for building the project, a breakout board for ESP8266 chip can be used. The breakout board used in this project is shown in Fig. 3. However, if a project-specific PCB is used, the breakout board is not required. ESP8266-12x breakout board is easily available at a low cost.

ESP-12E/F chip is soldered to the breakout board at the given pads, and header pins are also soldered to the breakout board. Header pins are spaced 2.54mm apart. ESP-12E/F module can be plugged into any general-purpose PCB.

There are also three SMD resistors on the breakout board (Fig. 3). On the obverse of the board, there is space for fixing HT7333A, a low-drop 3.3V voltage regulator. Alternately, an external 3.3V regulator may be used for powering the circuit.

Manual ESP-12E/F programmer

ESP-12x can boot up in three different modes, as shown in the table. We are interested in the first two boot-up modes.

As can be seen from the table, at boot-up, GPIO2 and Ch_PD/En pins should always be held high, while GPIO15 should be held low. While using the standalone ESP-12E/F chip, add 10k/12k pull-up resistors on Ch_PD and GPIO2 pins, and pull-down (10k/12k) resistors on GPIO15 pin. Note that, pull-up and pull-down resistors are prebuilt on the breakout board.

Reset pin and GPIO0 should nor-

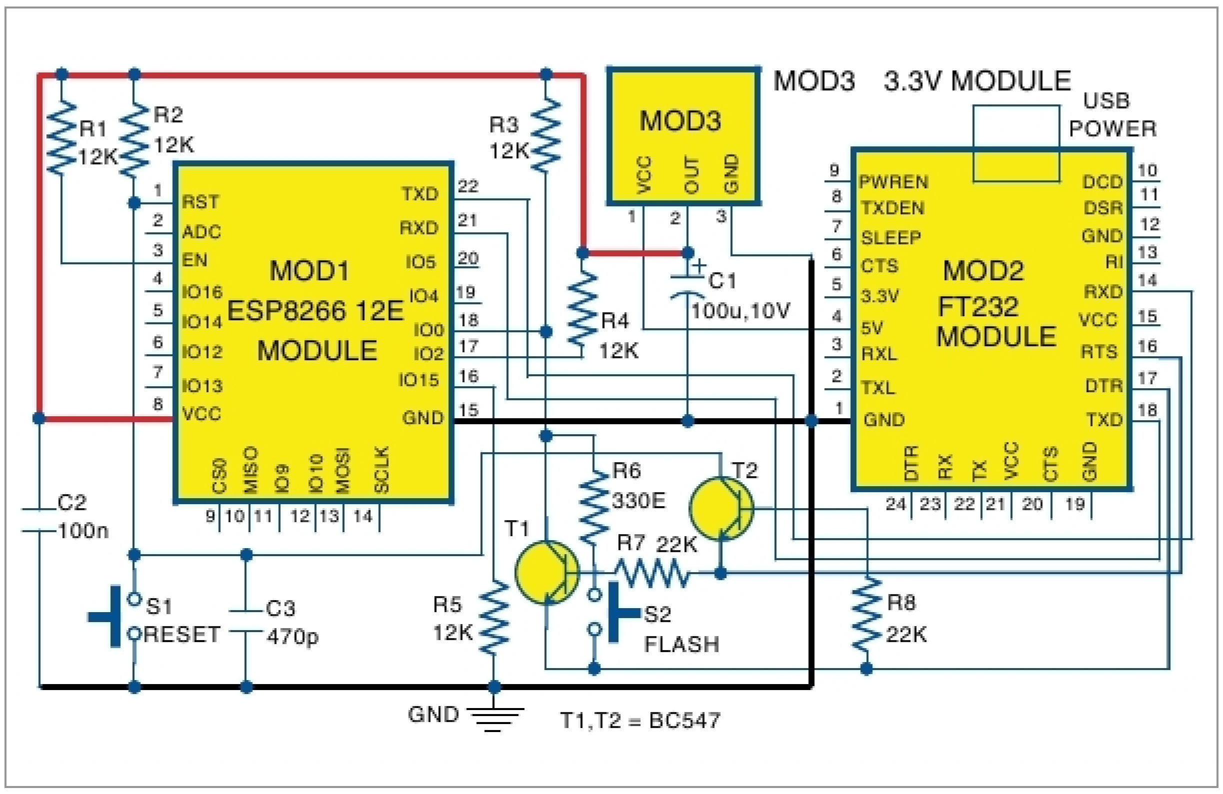

Fig. 6: Circuit diagram of the automatic ESP-12E/F programmer